

Accessibility Testing

Accessibility testing is a type of testing that is done to ensure that the application is usable by people with disabilities like hearing, poor sight, color blindness, etc. It is a type of usability testing. People with disabilities use assistive technology, which helps them use the product. It's a legal requirement in many countries. Eg: Speech recognition software, Screen reader software, Special keyboard made typically for those people, etc.

It has numbers of perks:

- Ensures equal access to information and features for all users. Disabled people aren't left out and can enjoy the product same like others.
- Avoids lawsuits or penalties. There are different laws in different countries that the product should be made accessible to every user, if this requirement is not met, then penalties are given as a punishment.
- Improves overall user experience and SEO (accessible sites are often more usable for everyone).
- Cater to market for disabled people. A software product can cater to this big market if it's made disabled-friendly. Accessibility issues in software can be resolved if Accessibility Testing is made part of the normal software testing life cycle.

Accessibility issues (WCAG basics)

Web Content Accessibility Guidelines (WCAG) aims to provide a single shared standard for web content accessibility that meets the needs of individuals, organizations, and governments internationally. The WCAG documents explain how to make web content more accessible to people with disabilities. Its main target are for:

- Web content developers (page authors, site designers, etc.)

- Web authoring tool developers
- Web accessibility evaluation tool developers
- Others who want or need a standard for web accessibility

WCAG is based on four principles:

- **Perceivable** - Users must be able to perceive content. Eg: Missing alt text for images
- **Operable** - Users must be able to navigate easily Eg: No keyboard navigation, inaccessible menus
- **Understandable** - Users must understand content and UI. Eg: Inconsistent labels, unclear error messages
- **Robust** - Content must work with assistive tech. Eg: Poor ARIA labeling, broken semantic HTML

Tools for Accessibility Testing

- **Screen Readers** Simulate how visually impaired users experience site (e.g., NVDA, VoiceOver, JAWS)

Tool	Platform	Description
NVDA (NonVisual Desktop Access)	Windows	Free, open-source screen reader widely used by developers
JAWS (Job Access With Speech)	Windows	Commercial tool with deep customization options
VoiceOver	macOS / iOS	Built into Apple devices; useful for testing mobile accessibility
TalkBack	Android	Built into Android; reads out UI elements and helps test mobile apps

- **Color Contrast Checkers** Test text/background contrast (e.g., WebAIM Contrast Checker, Chrome DevTools contrast)

Tool	Description
WebAIM Contrast Checker	Online checker that calculates text/background contrast ratio
Color Contrast Analyzer (TPGi)	Desktop app for designers & QA testers
Chrome DevTools → Accessibility → Contrast	Built-in contrast testing feature